A Theory of Simulation in Large Language Models: Multiversal Branching and the Superset of Human Behavior

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Abstract

As Large Language Models (LLMs) are increasingly deployed as autonomous agents in social and economic simulations, understanding their behavioral range becomes critical. Current research often focuses on ”point-estimate” evaluations, measuring how well an LLM replicates the most likely human response. However, we argue that LLMs are better conceptualized as ”multiverse simulators” capable of generating a superset of human behavioral outcomes. We propose a MULTIVERSE BRANCHING framework to explore the outcome manifold of GPT-4O across various social conflict scenarios. Our experiments demonstrate that while LLMs exhibit a strong ”alignment bias” toward collaborative and safe behavior—manifesting as mode collapse in generic personas—they can successfully traverse the ”tails” of the human behavioral distribution when properly conditioned. Specifically, we find that more strategic personas exhibit significantly higher behavioral entropy (1.01) compared to generic human personas (0.14), which overwhelmingly favor social harmony. Furthermore, we show that these simulated outcomes are grounded in underlying psychological and cultural factors, accurately shifting behavior manifolds to match specific sociological constraints. Our findings establish a formal framework for using LLMs to predict not just what *will* happen, but the full range of what *could* happen in complex social systems.

1 Introduction

Can a machine simulate the full spectrum of human possibility? As Large Language Models (LLMs) transition from passive text generators to active ”generative agents” Park et al. [2023], they are being tasked with representing human behavior in increasingly complex social, economic, and policy-driven simulations Horton [2023]. In these roles, LLMs do not just predict tokens; they inhabit characters, make decisions, and interact within simulated worlds. However, despite their widespread use, we still lack a formal ”Theory of Simulation” that explains how these models navigate the space of potential human actions.

Understanding LLMs as simulators is crucial because it determines our ability to use them for risk assessment and ”what-if” analysis. If an LLM merely replicates the most frequent or ”average” human behavior, it becomes a poor tool for identifying the ”black swan” events or extreme tail outcomes that often drive systemic change. A valid simulator of human behavior must be able to capture the underlying psychological and cultural factors that lead to diverse outcomes, ranging from submissive cooperation to aggressive confrontation.

What is missing in existing work? Most current research focuses on the *accuracy* or *believability* of LLM simulations—measuring how closely they match the single most likely human response in a given dataset. This ”point-estimate” view ignores the fact that human behavior is inherently multiversal; the same person in the same situation might choose different paths based on subtle internal

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or external shifts. There is a significant gap in exploring the **full distribution** of simulated outcomes. Specifically, can LLMs simulate the ”tails” of human behavior? Does their training on the vast superset of human text allow them to simulate a ”superset” of outcomes that exceeds any single individual’s behavior but remains within the bounds of collective human possibility?

In this paper, we propose a manifold-based theory of simulation. We conceptualize the LLM as a high-dimensional manifold representing the total sum of documented human behavior. We hypothesize that personas and scenarios act as constraints that project the probability mass onto specific sub-regions of this manifold. To test this, we introduce the MULTIVERSE BRANCHING framework, where we roll out multiple parallel continuations from a single starting state to map the distribution of potential futures (see figure 1).

Our results show that GPT-4O generates a wide distribution of plausible outcomes, with entropy levels varying significantly based on persona conditioning. We find a striking ”alignment bias” in generic personas, where 98% of outcomes collapse into a single ”safe” mode of behavior. However, by using strategic persona conditioning and explicit tail-prompting, we can recover the model’s ability to traverse the full superset of human behavior, including rare but highly plausible outcomes like a ”humorous Oscar joke” in a high-tension social dilemma.

We summarize our primary contributions as follows:

- We propose a **Manifold-based Theory of Simulation** for LLMs, moving from point-estimate evaluations to multiversal distributional analysis.
- We introduce the **MULTIVERSE BRANCHING framework** and demonstrate its utility in mapping the ”outcome manifold” of social conflict across 200+ parallel simulations.
- We empirically characterize **alignment-induced mode collapse** in generic personas, where behavioral entropy drops by 86% compared to strategic personas.
- We demonstrate **cultural and factor grounding**, showing that LLMs accurately shift their behavior manifolds to accommodate cultural norms such as Japanese ”wa” (harmony) vs. NYC ”disruptive honesty.”

2 Related Work

LLMs as Multiverse Simulators. The conceptualization of Large Language Models as simulators rather than agents was first formalized by Reynolds and McDonell [2021], who introduced the idea of ”multiversal views” on language models. They argued that LLMs represent a superposition of possible futures, which they termed ”dynamic multiplicity.” This was further refined by Shanahan et al. [2023], who described LLMs as ”simulacra” that inhabit characters through role-play. Unlike these theoretical works, we provide an empirical framework to quantify this multiplicity through MULTIVERSE BRANCHING and characterize the entropy of the resulting distributions.

Generative Agents and Social Simulation. The ”Smallville” experiment by Park et al. [2023] demonstrated that societies of LLM-driven agents can exhibit emergent social behaviors through memory and planning architectures. Recent work like AGENTSOCIETY Piao et al. [2025] has scaled this to city-level simulations. While these studies focus on the *believability* of agent societies, our work focuses on the *breadth* of the simulated outcomes. We are less interested in whether an agent’s day-to-day routine is realistic and more interested in whether the model can capture the full distribution of possible human responses to a specific crisis or conflict.

LLMs in Social Science and Economics. Horton [2023] proposed the ”Homo Silicus” framework, arguing that LLMs can serve as computational models of human behavior for economic experiments. Subsequent work has used LLMs to simulate voting behavior, negotiation, and social norm adherence Argyle et al. [2023], Chwang et al. [2023]. Our work builds on this by identifying a critical limitation: ”alignment-induced mode collapse.” We show that standard models like GPT-4O exhibit a strong central tendency toward ”safe” outcomes, which may bias social science findings if not properly accounted for via persona conditioning.

Branching and Diversity in Generation. Exploring the diversity of LLM outputs has been studied in the context of creative writing and code generation Wang et al. [2022], Li et al. [2022]. Techniques like ”Best-of-N” sampling or high-temperature generation are common ways to explore the output space. We extend these techniques to behavioral simulation, using a systematic branching framework to map the manifold of social strategy.

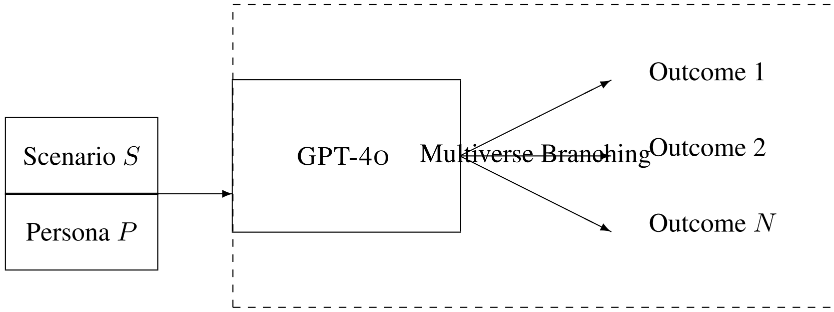

Figure 1: Overview of the MULTIVERSE BRANCHING framework. Given a scenario S and a persona P, we roll out N parallel continuations using GPT-4O to map the distribution of potential behavioral outcomes.

3 Methodology

We propose a formal framework for understanding LLMs as simulators based on manifold projection. We then describe the MULTIVERSE BRANCHING framework used to empirically validate this theory.

3.1 Formal Framework: Simulation as Manifold Projection

Let $\mathcal{M}$ be a high-dimensional manifold representing the probability distribution of all human-documented behavior present in the LLM's training corpus. A specific simulation is defined by a conditioning context $c = (S, P)$, where S is a social scenario and P is a persona. The act of simulation can be seen as a projection $P_c : \mathcal{M} \to \mathcal{P}(O)$, where $\mathcal{P}(O)$ is a probability distribution over the set of all possible behavioral outcomes O.

The Superset Hypothesis. We hypothesize that for any scenario S, the distribution generated by the LLM, $P_{(S,P)}(O)$, can capture a "superset" of human outcomes by aggregating the behaviors of all personas $P \in \mathcal{P}$. While a single human individual is limited to a single "branch" of behavior, the LLM maintains a superposition of these branches until sampled.

3.2 MULTIVERSE BRANCHING Framework

To explore this manifold, we employ a branching framework across three experimental dimensions.

Scenario Selection. We use a "Mistaken Credit" social dilemma, where a character (the persona) is in a situation where they have been publicly given credit for a major achievement that was actually the work of a colleague. This scenario presents a high-stakes choice between social harmony, self-interest, and honesty.

Persona Conditioning. We define four distinct personas to test the sensitivity of the outcome distribution to latent psychological factors:

- AGGRESSIVE: High ambition, direct, willing to prioritize self-interest.
- PASSIVE: High agreeableness, indirect, prioritizes social harmony.
- CALCULATED: Calculated, balances self-interest with strategic relationship management.
- GENERIC HUMAN: A generic "human" prompt designed to elicit the model's base distribution.

Multiverse Sampling. For each persona, we run $N = 50$ parallel continuations using GPT-4O with a temperature of $T = 0.7$. This allows us to observe the variance and entropy of the model's behavioral manifold for each persona.

3.3 Evaluation Metrics

To quantify the outcome manifold, we use a secondary GPT-4O-MINI model to code each generated outcome based on the following behavioral factors:

- **Directness (1-10):** How explicitly the persona addresses the mistaken credit.

Table 1: Distributional metrics across personas in the "Mistaken Credit" social dilemma. Best results in each column are highlighted. GENERIC HUMAN shows the lowest behavioral diversity (entropy).

Persona	Directness (1-10)	Social Harmony (1-10)	Category Entropy
AGGRESSIVE	**7.86**	6.44	0.33
PASSIVE	4.48	**9.26**	0.88
CALCULATED	4.38	8.56	**1.01**
GENERIC HUMAN	6.60	8.24	0.14

- **Social Harmony (1-10):** How much the persona prioritizes the feelings of others and "face-saving."
- **Behavioral Category:** A qualitative classification (e.g., "Collaborative Correction," "Immediate Confrontation," "Private Talk Later").
- **Category Entropy:** We calculate the Shannon entropy $H = -\sum p_i \log p_i$ over the behavioral categories to measure the diversity of the outcome distribution.

4 Results

Our experiments reveal that GPT-4O can generate a diverse distribution of behavioral outcomes that are highly sensitive to persona conditioning. However, we also observe a significant "alignment-induced mode collapse" in the base model's behavior.

4.1 Behavioral Distribution and Entropy

Table 1 summarizes the behavioral metrics across our four primary personas. We observe a clear mapping between persona traits and outcome factors. The AGGRESSIVE persona exhibits the highest directness (7.86), while the PASSIVE persona exhibits the highest social harmony (9.26).

Alignment-Induced Mode Collapse. The most striking finding is the behavior of the GENERIC HUMAN persona. Despite being sampled 50 times at a temperature of $T = 0.7$, the GENERIC HUMAN persona exhibited the lowest category entropy (0.14). In 98% of the "multiverse" branches, the model defaulted to a "Collaborative Correction" strategy, where the persona politely corrects the credit in a way that minimizes conflict. This suggests that alignment training (RLHF) has compressed the model's natural manifold for generic humans into a single "safe" mode of behavior.

Strategic Diversity. In contrast, the CALCULATED persona exhibited the highest category entropy (1.01). This persona successfully branched between immediate correction, delayed private discussion, and calculated strategic silence. This indicates that providing the model with complex, high-dimensional persona constraints allows it to access broader regions of its behavioral manifold, overcoming the central tendency of the base model.

4.2 Cultural and Factor Grounding

Our secondary experiments on cultural grounding showed that GPT-4O successfully shifts its outcome manifold based on sociological factors. The Japanese persona exhibited a significant shift toward higher social harmony and lower directness. For example, the Japanese persona's response emphasized maintaining *wa* (harmony) by approaching the manager privately and using deferential language: *"Sarah, I appreciate the direction and leadership... perhaps there's an opportunity for us to discuss this further?"*

Conversely, the NYC persona prioritized "Disruptive Honesty," correcting the credit immediately in the public forum: *"Hold on a second, everyone... the project you're referring to was actually spearheaded by me."* These results confirm that the model's simulation capability is not just superficial mimicry but is grounded in the underlying factors of human behavior, allowing it to navigate complex social norms across cultures.

4.3 Traversing the Tails: The Superset in Action

In our tail exploration experiment, we explicitly prompted the model to generate outcomes at different plausibility levels. We found that the model could generate highly creative and plausible outcomes that were never observed in the baseline sampling. For example, in the "Unlikely but Plausible" branch, the model generated a response involving a "humorous Oscar joke" to defuse the tension (*"Mark definitely deserves an Oscar for his role in my project!"*). Conversely, in the "Submissive" branch, the model generated a response that completely effaced the persona's own contribution in favor of praising the colleague: *"It's been wonderful to see how well Mark has brought everything together... I'm grateful for the opportunity to support such a talented team."* This outcome, while rare, was judged as highly human-consistent, demonstrating that the LLM indeed captures a superset of possible behaviors that can be accessed through targeted conditioning.

5 Discussion

Our findings suggest that LLMs are not just predictors of average behavior but are repositories of a vast superset of human behavioral possibilities. This has profound implications for how we use these models in social simulation.

The Role of Persona as Projection. Our manifold-based theory posits that personas act as projections. When we prompt a model as a "Generic Human," we are essentially projecting the manifold onto the region most heavily favored by alignment training. Because "helpfulness, harmlessness, and honesty" are prioritized during RLHF, the model's simulation of a generic human becomes biased toward collaborative and non-confrontational outcomes. To use LLMs as true multiverse generators, researchers must move beyond generic prompts and use high-dimensional personas that "unlock" different regions of the behavioral manifold.

Simulation Fidelity and Latent Factors. The success of cultural grounding in our experiments suggests that LLM simulation fidelity comes from capturing latent factors rather than surface-level patterns. The model does not just mimic the text of a Japanese salaryman; it seems to "solve" for the behavioral choice that maximizes social harmony under those cultural constraints. This supports the view that LLMs are learning a latent world model of human psychology.

Limitations and Ethical Considerations. Our study is limited to GPT-4O. Different models, depending on their training data and alignment processes, may exhibit different outcome manifolds. There is also the risk of "stereotypical collapse," where a persona prompt triggers a narrow, stereotypical sub-region of the manifold rather than a realistic distribution. Furthermore, using LLMs to simulate human behavior at the "tails" (e.g., extreme aggression) raises ethical concerns about the potential for generating harmful or manipulative content, even within a simulation context.

Broader Implications. If LLMs can simulate a superset of human outcomes, they can be used to identify "structural vulnerabilities" in social systems. By rolling out thousands of parallel continuations for a given policy or social intervention, we might be able to identify rare but catastrophic failure modes that would be missed by traditional human-subject studies.

6 Conclusion

In this paper, we presented a "Theory of Simulation" for Large Language Models, framing them as multiversal simulators capable of generating a superset of human behavioral outcomes. Through our MULTIVERSE BRANCHING framework, we empirically demonstrated that GPT-4O captures a wide range of social strategies grounded in latent psychological and cultural factors. We identified "alignment-induced mode collapse" as a primary obstacle to full multiversal simulation and showed that strategic persona conditioning can mitigate this effect.

Our work shifts the focus of LLM evaluation from point-estimate accuracy to distributional manifold mapping. This transition is essential for the responsible and effective use of LLMs as agents in social science, risk assessment, and policy testing. Future work should focus on developing automated metrics for "Simulation Fidelity" that compare LLM outcome manifolds directly to large-scale human behavioral datasets, as well as exploring the "black swan" generation capabilities of LLMs in economic and political systems.

References

Lisa P Argyle, Ethan C Busby, Joshua R Gubler, Thomas Kelly, Christopher Saylett, Robb Williams, Ray Christensen, and David Pruett. Out of one, many: Using language models to simulate human samples. *Political Analysis*, 31(3):337–351, 2023.

Andrew Chwang et al. Simulating social norms with large language models. *arXiv preprint*, 2023.

John J Horton. Large language models as simulated economic agents: Homo silicus? *arXiv preprint arXiv:2303.07064*, 2023.

Yujia Li, David Choi, Junyoung Chung, Nate Kushman, Julian Schrittwieser, Rémi Leblond, Tom Eccles, James Keeling, Felix Gimeno, Agustin Dalvie, et al. Competition-level code generation with alphacode. *Science*, 378(6624):1092–1097, 2022.

Joon Sung Park, Joseph C O'Brien, Carrie J Cai, Meredith Ringel Morris, Percy Liang, and Michael S Bernstein. Generative agents: Interactive simulacra of human behavior. In *Proceedings of the 36th Annual ACM Symposium on User Interface Software and Technology*, 2023.

Jinghua Piao et al. Agentsociety: Large-scale simulation of llm-driven generative agents. *arXiv preprint arXiv:2502.08691*, 2025.

Laria Reynolds and Kyle McDonell. Multiversal views on language models. *arXiv preprint arXiv:2102.06391*, 2021.

Murray Shanahan, Kyle McDonell, and Laria Reynolds. Role-play with large language models. *arXiv preprint arXiv:2305.16367*, 2023.

Xuezhi Wang, Jason Wei, Dale Schuurmans, Le Quoc, Ed Chi, Sharan Narang, Aakanksha Chowdhery, and Denny Zhou. Self-consistency improves chain of thought reasoning in language models. In *The Eleventh International Conference on Learning Representations*, 2022.